

1. What is an Artificial Neural Network (ANN)?

Definition

An Artificial Neural Network (ANN) is a computational model inspired by the human brain that learns patterns from data and makes predictions.

ANN is made up of:

Neurons
↓
Layers
↓
Connections (Weights)

Why ANN?

Suppose you want to predict:

Student Performance

Inputs:

Study Hours Attendance Assignments

5 90% 8

Output:

Pass

Instead of writing hundreds of rules manually, ANN learns the relationship automatically.

2. Biological Neuron vs Artificial Neuron

Human Brain Neuron

Structure:

Dendrites
↓
Cell Body
↓
Axon

Dendrites

Receive signals from other neurons.

Cell Body

Processes information.

Axon

Sends output signal.

Artificial Neuron

Equivalent structure:

```
Inputs
  ↓
Weights
  ↓
Summation
  ↓
Activation Function
  ↓
Output
```

Comparison

Human Brain	ANN
Dendrites	Inputs
Synapse	Weights
Cell Body	Summation
Activation	Activation Function
Axon	Output

3. What is a Neuron?

Neuron is the smallest processing unit of a neural network.

Example:

```
x1 = Study Hours
x2 = Attendance
x3 = Assignments
```

Neuron receives these inputs and produces an output.

Mathematical Representation

First calculate weighted sum:

$$z = w_1x_1 + w_2x_2 + w_3x_3 + bz = w_1x_1 + w_2x_2 + w_3x_3 + bz = w_1x_1 + w_2x_2 + w_3x_3 + b$$

Then:

$$y = f(z)$$

Where:

- x = Inputs
 - w = Weights
 - b = Bias
 - f = Activation Function
 - y = Output
-

4. Inputs (Features)

Inputs are variables used for prediction.

Example 1: House Price Prediction

Inputs:

Area
Bedrooms
Age
Location Score

Output:

House Price

Example 2: Student Performance

Inputs:

Study Hours
Attendance
Assignments

Output:

Pass or Fail

5. Weights

Definition

Weights indicate the importance of each input.

Example:

Student Prediction

Feature	Weight
Study Hours	0.8
Attendance	0.5
Assignments	0.2

Interpretation:

Study Hours

has the strongest influence.

Why Weights?

Without weights:

All inputs are equally important.

With weights:

Important features contribute more.

Numerical Example

Inputs:

x1 = 2
x2 = 4

Weights:

w1 = 3
w2 = 5

Calculation:

$(2 \times 3) + (4 \times 5)$
 $6 + 20$
26

6. Bias

Definition

Bias is an additional parameter that shifts the output.

Formula:

$$z = \sum wx + bz = \sum wx + b$$

Example

Weighted Sum:

26

Bias:

b = 4

Output:

30

Why Bias is Needed?

Without bias:

Output always starts from zero.

Bias allows:

Flexible learning.

7. Weighted Sum Calculation

Example:

Inputs:

$x_1 = 3$
 $x_2 = 4$

Weights:

$w_1 = 2$
 $w_2 = 5$

Bias:

1

Calculation:

$z = (3 \times 2) + (4 \times 5) + 1$
 $= 6 + 20 + 1$
 $= 27$

8. Activation Function (Introduction)

After weighted sum:

$z = 27$

Apply activation function.

Purpose:

Introduce non-linearity.

Without Activation:

Network becomes:

Linear Regression

No matter how many layers exist.

Popular Activations:

Sigmoid

Output:

0 to 1

Tanh

Output:

-1 to 1

ReLU

Output:

0 to ∞

Most commonly used.

Softmax

Used for:

Multi-Class Classification

9. Layers in ANN

ANN is composed of layers.

Input Layer

Receives features.

Example:

Age
Salary
Experience

Three inputs:

3 neurons

Hidden Layer

Learns patterns.

Example:

Input Layer
↓
Hidden Layer
↓
Output Layer

Why Hidden?

Because users cannot directly see its computations.

Output Layer

Produces final prediction.

Examples:

Binary Classification

Yes
No

Multi-Class Classification

Cat
Dog
Bird

Regression

House Price

10. ANN Architecture

Simple ANN:

Input Layer
↓
Hidden Layer
↓
Output Layer

Deep ANN:

Input Layer
↓
Hidden Layer 1
↓
Hidden Layer 2
↓
Hidden Layer 3
↓
Output Layer

Example

Image Recognition

Layer 1 learns:

Edges

Layer 2 learns:

Shapes

Layer 3 learns:

Objects

11. How ANN Works

Complete Flow:

Input Data
↓
Weights Applied
↓
Weighted Sum
↓

Bias Added
↓
Activation Function
↓
Output

Example

Input:

```
Study Hours = 5  
Attendance = 90  
Assignments = 8
```

↓

ANN Processing

↓

Output:

Pass

12. ANN Code Example (TensorFlow/Keras)

Step 1: Import Libraries

```
import numpy as np  
from tensorflow.keras.models import Sequential  
from tensorflow.keras.layers import Dense
```

Step 2: Sample Data

```
X = np.array([  
    [2, 80],  
    [4, 85],  
    [6, 90],  
    [8, 95]  
)  
  
y = np.array([0, 0, 1, 1])
```

Explanation:

Feature 1 = Study Hours
Feature 2 = Attendance

Target:
0 = Fail
1 = Pass

Step 3: Create ANN

```
model = Sequential()  
  
model.add(Dense(  
    units=4,  
    activation='relu',  
    input_shape=(2,) ) )  
  
model.add(Dense(  
    units=1,  
    activation='sigmoid'  
))
```

Explanation

Dense Layer

Dense(4)

means:

4 neurons in hidden layer

Activation='relu'

Uses ReLU activation.

Input Shape

input_shape=(2,)

means:

2 input features

Output Layer

Dense(1, activation='sigmoid')

means:

1 neuron

Binary Classification:

Pass / Fail

Step 4: Compile Model

```
model.compile(  
    optimizer='adam',  
    loss='binary_crossentropy',  
    metrics=['accuracy']  
)
```

Optimizer

adam

Updates weights automatically.

Loss Function

binary_crossentropy

Used for binary classification.

Metrics

accuracy

Measures correctness.

Step 5: Train Model

```
model.fit(  
    X,  
    Y,  
    epochs=50,  
    verbose=1  
)
```

Epochs

50

means:

Dataset processed 50 times.

Step 6: Prediction

```
prediction = model.predict(  
    np.array([[7, 92]])  
)  
  
print(prediction)
```

Output:

```
[[0.93]]
```

Meaning:

93% probability of Pass

ANN Summary

```
Input Features  
    ↓  
Weights  
    ↓  
Weighted Sum  
    ↓  
Bias  
    ↓  
Activation Function  
    ↓  
Hidden Layer  
    ↓  
Output Layer  
    ↓  
Prediction
```

Interview Questions

What is ANN?

ANN is a neural network inspired by the human brain that learns patterns from data.

What are Weights?

Weights determine the importance of input features.

What is Bias?

Bias shifts the output and improves model flexibility.

What is a Hidden Layer?

A layer between input and output that learns patterns from data.

Why do we need Activation Functions?

To introduce non-linearity so the network can learn complex relationships.

Difference between ANN and Deep Neural Network?

- ANN → One or few hidden layers.
- Deep Neural Network (DNN) → Multiple hidden layers.

-
1. Artificial Neurons
 2. Inputs
 3. Weights
 4. Bias
 5. Weighted Sum
 6. Activation Functions
 7. Input Layer
 8. Hidden Layer
 9. Output Layer
 10. ANN Architecture
 11. ANN Workflow
 12. ANN Implementation in TensorFlow/Keras

we will understand:

```
Input
↓
Prediction
↓
Error
↓
Weight Update
↓
Better Prediction
```

This learning process is the heart of Deep Learning.

1. Why Does a Neural Network Need Learning?

Imagine a student appearing for a test.

Question:

$2 + 3 = ?$

Student Answer:

4

Correct Answer:

5

Error:

1

The student identifies the mistake and improves.

A Neural Network learns exactly the same way.

Neural Network Learning Process

```
Input
↓
Prediction
↓
Compare with Actual Answer
↓
Calculate Error
↓
Update Weights
↓
Generate Better Prediction
```

This process repeats thousands of times until the error becomes very small.

2. Forward Propagation

Definition

Forward Propagation is the process of moving information from the Input Layer to the Output Layer to generate predictions.

Why is it called Forward?

Because data moves:

```
Input Layer
  ↓
Hidden Layer
  ↓
Output Layer
```

Only in the forward direction.

Example

Suppose we want to predict whether a student will pass.

Inputs:

Feature	Value
Study Hours	6
Attendance	90

The network receives these values and calculates an output.

Step-by-Step Forward Propagation

Step 1: Input Layer

Inputs enter the network.

```
x1 = 6
x2 = 90
```

Step 2: Multiply by Weights

Each input has an associated weight.

```
w1 = 0.8
w2 = 0.4
```


Calculation:

$$\begin{aligned} &(6 \times 0.8) \\ &+ \\ &(90 \times 0.4) \end{aligned}$$

Step 3: Add Bias

Bias helps shift predictions.

$$+ b$$

Suppose:

$$b = 1$$

Step 4: Apply Activation Function

Activation determines whether the neuron should activate.

Example:

ReLU

$$\text{ReLU}(x) = \max(0, x)$$

Step 5: Generate Output

Output might be:

$$0.95$$

Meaning:

95% probability of passing.

Complete Forward Propagation Flow

```
Inputs
↓
Weights
↓
Weighted Sum
↓
Bias
↓
Activation Function
```

↓
Prediction

Real-Life Analogy

Suppose a bank decides whether to approve a loan.

Inputs:

Salary
Age
Credit Score

The bank combines these factors and makes a decision.

ANN does exactly the same thing using mathematical calculations.

3. What Happens After Prediction?

Suppose:

Actual Result:

Pass

Network Prediction:

Fail

The prediction is wrong.

Question:

How does the network know it made a mistake?

Answer:

Loss Function

4. Loss Function

Definition

A Loss Function measures how wrong the network's prediction is.

Why Do We Need Loss?

Without loss:

Network cannot know if prediction is good or bad.

Loss acts like an exam score.

Example

Actual House Price:

500000

Predicted Price:

450000

Difference:

50000

Loss measures this error.

Characteristics of a Good Loss Function

A good loss function should:

Small Error

Produce small loss.

Large Error

Produce large loss.

Easy to Optimize

Allow weight updates efficiently.

5. Mean Squared Error (MSE)

Most common loss function for regression.

Used when output is numerical.

Examples:

- House Price Prediction
 - Temperature Prediction
 - Sales Forecasting
-

Theory

MSE calculates:

Actual - Predicted

Then squares the difference.

Why square?

Because:

Negative errors become positive.

Example:

$$-10^2 = 100$$

$$10^2 = 100$$

No cancellation occurs.

Interpretation

Small MSE

Good model.

Large MSE

Poor model.

6. Mean Absolute Error (MAE)

Another popular regression loss.

Instead of squaring:

Take absolute value.

Example

Actual:

100

Prediction:

90

Error:

10

MAE:

10

Difference Between MSE and MAE

MSE	MAE
Squares Error	Uses Absolute Error
Sensitive to Outliers	Less Sensitive
Punishes Large Errors More	Treats Errors Equally

7. Binary Cross Entropy

Used for:

Binary Classification

Examples:

- Pass / Fail
- Spam / Not Spam
- Yes / No

Theory

If the network predicts:

Pass = 99%

and student actually passes,

loss is very small.

If network predicts:

Pass = 1%

but student passes,

loss becomes very large.

Binary Cross Entropy measures this mismatch.

8. Cost Function

Students often confuse:

Loss Function

and

Cost Function

Loss

Error for one training example.

Example:

Student 1 prediction error.

Cost

Average error of entire dataset.

Example:

1000 students.

Average error of all predictions.

Goal

Minimize Cost Function

because lower cost means better model performance.

9. Why Do We Need Backpropagation?

Suppose:

Actual:

100

Prediction:

80

Error:

20

Question:

How can we reduce this error?

Answer:

10. What is Backpropagation?

Definition

Backpropagation is the process of sending error information backward through the network and updating weights.

Why Backpropagation?

Weights determine predictions.

Wrong prediction means:

Weights are not optimal.

Backpropagation adjusts them.

Real-Life Analogy

Suppose a student scores:

40/100

Teacher says:

You made mistakes in Algebra.

Student studies Algebra and improves.

Similarly:

Backpropagation identifies bad weights.

Then updates them.

Backpropagation Flow

Prediction

↓

Loss Calculation
↓
Find Error Contribution
↓
Update Weights
↓
Reduce Error

11. Gradient

Definition

Gradient measures how much the loss changes when weights change.

Simple Meaning

Gradient tells:

Which direction increases error?

Neural Network moves in the opposite direction.

Mountain Analogy

Imagine standing on a mountain.

Goal:

Reach the lowest point.

Gradient tells:

Which way is uphill?

Move opposite direction.

12. Gradient Descent

Definition

Gradient Descent is an optimization algorithm used to reduce loss.

Goal

Find Best Weights
that produce minimum error.

Process

Current Weight
↓
Calculate Gradient
↓
Adjust Weight
↓
Lower Loss

Repeat until convergence.

13. Learning Rate

Definition

Learning Rate controls how much weights change.

Symbol:

η (eta)

Small Learning Rate

Advantages:

- Stable learning

Disadvantages:

- Slow training
-

Large Learning Rate

Advantages:

- Faster learning

Disadvantages:

- May overshoot optimum solution

Common Values

0.1
0.01
0.001

14. Epoch

Definition

One complete pass through the training dataset.

Example:

Dataset:

1000 records

After processing all 1000 records once:

1 Epoch

Example:

epochs = 50

Means:

Dataset processed 50 times.

Why Multiple Epochs?

One pass is usually not enough.

Repeated learning improves weights.

15. Batch Size

Instead of training on:

`1000 rows`

all at once,

split into smaller groups.

Example:

`32 rows`

at a time.

Benefits:

- Faster training
 - Lower memory consumption
 - Better optimization
-

16. Iteration

Definition

One batch processed during training.

Example

Dataset:

`1000 rows`

Batch Size:

`100`

Iterations:

```
1000 / 100  
=  
10
```

Relationship:

```
1 Epoch  
↓  
10 Iterations
```

17. Complete Learning Cycle

This is what happens during training:

```
Input Data  
↓  
Forward Propagation  
↓  
Prediction  
↓  
Loss Function  
↓  
Calculate Error  
↓  
Backpropagation  
↓  
Gradient Descent  
↓  
Update Weights  
↓  
Repeat
```

18. ANN Training Code

```
import numpy as np  
from tensorflow.keras.models import Sequential  
from tensorflow.keras.layers import Dense  
  
X = np.array([  
    [2,80],  
    [4,85],  
    [6,90],  
    [8,95]  
)  
  
y = np.array([0,0,1,1])  
  
model = Sequential([  
    Dense(8, activation='relu', input_shape=(2,)),  
    Dense(4, activation='relu'),
```

```
Dense(1, activation='sigmoid')
])

model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)

model.fit(
    X,
    Y,
    epochs=100,
    batch_size=2
)

model.summary()
```

What Happens Internally During Training?

```
Epoch 1
↓
Forward Propagation
↓
Loss
↓
Backpropagation
↓
Weight Update

Epoch 2
↓
Forward Propagation
↓
Loss
↓
Backpropagation
↓
Weight Update

...
```

Until loss becomes very small.

Most Important Interview Questions

What is Forward Propagation?

Passing inputs through the network to generate predictions.

What is a Loss Function?

A function that measures prediction error.

Difference Between Loss and Cost?

Loss = one example error.

Cost = average dataset error.

What is Backpropagation?

Process of updating weights using prediction errors.

What is Gradient Descent?

Optimization algorithm used to minimize loss.

What is Learning Rate?

Controls the size of weight updates.

What is an Epoch?

One complete pass through training data.

What is Batch Size?

Number of samples processed before updating weights.

What is an Iteration?

One batch processed during training.

clearly understand:

- How ANN generates predictions (Forward Propagation)
- How prediction error is measured (Loss Functions)
- Difference between MSE, MAE, and Binary Cross Entropy
- Loss vs Cost Function
- How networks learn (Backpropagation)
- Gradient and Gradient Descent
- Learning Rate
- Epochs, Batch Size, and Iterations
- Complete ANN training cycle